

4- §. Grin funksiyasi usuli

Elliptik tipdagi tenglamalar uchun chegaraviy masalalarni yechishda (Manba funksiya) Grin funksiyasi muhim o'rin tutadi. Bo'laklari silliq S sirt bilan chegaralangan D sohani qaraylik.

4.1-ta'rif. (4.2) Laplas tenglamasi uchun D sohada ichki (tashqi) Dirixle masalasining **Grin funksiyasi** deb, ikkita $x, \xi \in D \cup S$ nuqtalarning funksiyasi bo'lgan va quyidagi shartlarni qanoatlantiruvchi $G(x, \xi)$ funksiyaga aytiladi [15]:

1) $G(x, \xi)$ funksiya ushbu ko'rinishga ega

$$G(x, \xi) = \frac{1}{4\pi} [E(x, \xi) + g(x, \xi)], \quad (4.56)$$

bu yerda $E(x, \xi)$ - Laplas tenglamasining (4.9) formula bilan aniqlangan fundamental (elementar) yechimi, $g(x, \xi)$ - funksiya esa $x \in D$ bo'yicha ham, $\xi \in D$ bo'yicha ham garmonik funksiya bo'lib, x bo'yicha har bir ξ lar uchun $\bar{D}$ sohada uzluksiz.

2) x, ξ nuqtalarning hech bo'lmaganda bittasi S ga tegishli bo'lsa,

$$G(x, \xi) = 0 \quad (4.57)$$

bo'ladi. Agar D soha chegaralanmagan bo'lsa, u holda $|x| \rightarrow \infty$ da $g(x, \xi) \rightarrow 0$ shart bajarilishi talab qilinadi.

Bu ta'rifga asosan, $G(x, \xi)$ funksiya ξ nuqtadan tashqari barcha D sohada garmonik bo'lib, $g(x, \xi)$ funksiya yordamida aniqlanadi. $g(x, \xi)$ funksiya esa o'z navbatida D sohada garmonik bo'lib, S da $-\ln(1/r)$, $n=2$ yoki $-r^{2-n}$, $n>2$ qiymatlarga teng. Bu yerda $g(x, \xi)$ shunday garmonik funksiyaki, u chegarada maxsus qiymatlarni qabul qiladi. Ayrim hollarda bunday funksiyani topish ancha qulay bo'ladi. Bu funksiya ma'lum bo'lgandan so'ng, chegarada ixtiyoriy qiymatni qabul qiluvchi garmonik funksiyani topish mumkin bo'ladi.

Grin funksiyasining xossalari:

1) $D \cup S$ da $G(x, \xi) \geq 0$;

2) agar D sohaning S chegarasi yetarlicha silliq sirt bo'lsa, u holda $G(x, \xi)$ funksiya S chegarada har bir $\xi \in D$ da to'g'ri normal

$\frac{\partial G(x, \xi)}{\partial n_x}$ hosilaga ega;

3) $G(x, \xi)$ funksiya $x \in D$ va $\xi \in D$ nuqtalarga nisbatan simmetrik funksiyaadir, ya'ni $G(x, \xi) = G(\xi, x)$;

4) $x \in D$ nuqtalarda quyidagi tenglik

$$-\frac{1}{(n-2)|S_1|} \int_S \frac{\partial G(x, \xi)}{\partial n_\xi} d_\xi S = 1$$

o'rinli. $|S_1|$ — birlik sferaning yuzi.

Agar $w = w(z)$ funksiya D ($n=2$ bo'lganda) sohani birlik doiraga (ya'ni $|w| < 1$ ga) konform akslantiruvchi funksiya bo'lsa, u holda Grin funksiyasi quyidagi formuladan

$$G(z, \zeta) = \frac{1}{2\pi} \ln \frac{1}{|w(z, \zeta)|}, \quad w(z, \zeta) = \frac{w(z) - w(\zeta)}{1 - \overline{w(z)} \cdot w(\zeta)},$$

$z = x + iy$, $\zeta = \xi + i\eta$ topiladi.

Grin funksiyasini tuzish. Qaralayotgan ko'pgina sohalarida Grin funksiyasi akslantirish usulidan foydalanib tuziladi [5], [10], [19].

1- masala. (4.2) Laplas tenglamasi uchun sharda ichki Dirixle masalasining Grin funksiyasini tuzing.

Yechish. Akslantirish usuli yordamida sharda Grin funksiyasini tuzamiz. U_R soha markazi koordinata boshida va radiusi R ga teng bo'lgan shar bo'lsin. Uni chegaralab turgan sferani S_R orqali belgilab olamiz. Faraz qilaylik, $y \neq 0$ bo'lib, y ga S_R sferaga nisbatan simmetrik nuqta y^* bo'lsin, ya'ni

$$|y||y^*| = R^2. \quad (4.58)$$

$|y| < R$ bo'lganligi uchun, y^* nuqta S_R sferadan tashqarida yotadi. Bu nuqta uchun ushbu inversiya almashtirish o'rinlidir:

$$y = \frac{R^2}{|y^*|^2} y^*, \quad y^* = \frac{R^2}{|y|^2} y, \quad |y| = \sqrt{y_1^2 + y_2^2 + y_3^2},$$

$$\text{yoki} \quad y_i^* = \frac{R^2}{|y|^2} y_i, \quad y_i = \frac{R^2}{|y^*|^2} y_i^*, \quad i = 1, 2, 3. \quad (4.59)$$

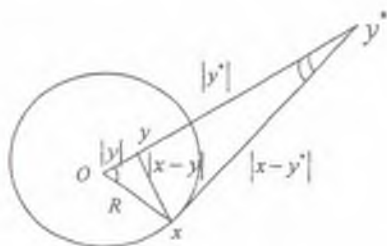
Grin funksiyani quyidagi ko'rinishda izlaymiz:

$$G(x, y) = v(x, y) + g(x, y), \quad (4.60)$$

bu yerda $v(x, y) = 1/4\pi|x-y|$ funksiya Laplas tenglamasining fundamental yechimi bo'lib, bu funksiya $y \neq x$ bo'lganda y bo'yicha ham, x bo'yicha ham Laplas tenglamasini qanoatlantiradi.

$g(x, y) = -\frac{\alpha}{4\pi|x-y^*|}$ funksiya esa U_R da y bo'yicha ham,

x bo'yicha ham garmonik funksiya bo'lib, $C^\infty(U_R)$ sinfga tegishlidir, ya'ni bu funksiyaning maxraji nolga aylanmaydi, α - keyinchalik aniqlanadigan son.



4.1 - chizma

Agar $x \in S_R$ bo'lsa, u holda 4.1 - chizmaga ko'ra Oxy^* va Oxy uchburchaklar o'xshash bo'ladi, chunki ular umumiy O burchakka ega va bu burchakni hosil qilgan tomonlari (4.58) va (4.59) tengliklarga ko'ra proporsionaldir, ya'ni

$$\frac{|y^*|}{R} = \frac{R}{y},$$

bu yerda $|y^*|$ va R - Oxy^* va Oxy uchburchaklarning mos ravishda tomonlarining uzunliklari.

Shunday qilib, uchburchaklarning o'xshashligidan

$$R \cdot |x-y| = |y| \cdot |x-y^*| \quad \text{yoki} \quad R \cdot |x-y| = |x-y^*| \cdot |y|$$

(4.61)

tengliklarni hosil qilamiz. Grin funksiyasining ta'rifiga ko'ra α sonini shunday tanlaymizki, $G(x, y)|_{S_R} = 0$ bo'lsin. Bundan va (4.60) dan

$$\alpha \cdot |x-y| = |x-y^*| \Rightarrow \alpha = \frac{|x-y^*|}{|x-y|}, \quad x \in S_R \quad (4.62)$$

kelib chiqadi. (4.61) ga asosan esa

$$\alpha = R/|y| \quad (4.63)$$

ega bo'lamiz.

Shunday qilib, (4.60) va (4.63)ga ko'ra

$$G(x, y) = \frac{1}{4\pi|x-y|} - \frac{R}{4\pi|y||x-y^*|} \quad (4.64)$$

ni hosil qilamiz.

(4.64) formulaning ikkinchi qo'shiluvchisini (4.59) ga asosan quyidagicha ifodalab olamiz:

$$\begin{aligned} \frac{R}{4\pi|y||x-y^*|} &= \frac{R}{4\pi|y||x-y^*|} \cdot \frac{|y|}{|y|} = \frac{R|y|}{4\pi|y|^2|x-y^*|} = \\ &= \frac{R|y|}{4\pi|x||y|^2 - y^*|y|^2|} = \frac{R|y|}{4\pi|x||y|^2 - R^2y^*}. \end{aligned} \quad (4.65)$$

(4.65) ifodani (4.64) formulaga qo'yib, shar uchun Grin funksiyasini quyidagi ko'rinishda topamiz:

$$G(x, y) = \frac{1}{4\pi|x-y|} - \frac{R|y|}{4\pi|x||y|^2 - R^2y^*} \quad (4.66)$$

2-masala. $\Delta u(x, y, z) = 0$ tenglamaning $D_1: z > 0$ yarim

fazodagi Grin funksiyasi $G(x, y, z; \xi, \eta, \zeta) = \frac{1}{4\pi} \left(\frac{1}{r} - \frac{1}{r_1} \right)$ ko'rinishga

ega bo'lishini ko'rsating, bu yerda

$$\begin{aligned} r &= \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}, \\ r_1 &= \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z+\zeta)^2} \end{aligned} \quad (4.2\text{-chizma}).$$

Yechish. (4.56) va (4.11) asosan, Grin funksiyasi

$$G(x, y, z; \xi, \eta) = \frac{1}{4\pi} \left(\frac{1}{r} + g(x, y, z; \xi, \eta, \zeta) \right)$$

ko'rinishga ega bo'ladi. $g(x, y, z; \xi, \eta, \zeta)$ funksiya, qaralayotgan sohada harmonik funksiya bo'lgani uchun $\Delta g(x, y, z) = 0$ va $\Delta g(\xi, \eta, \zeta) = 0$ tenglamalarni qanoatlantiradi, shu bilan birga (4.57)

shartga asosan (x, y, z) va (ξ, η, ζ) nuqtalarning hech bo'lmaganda bittasi ∂D_1 ga tegishli bo'lsa, $G(x, y, z, \xi, \eta, \zeta) = 0$, ya'ni $g(x, y, z; \xi, \eta, \zeta) = -1/r$ bo'lishi kerak. Demak, (x, y, z) va (ξ, η, ζ) nuqtalarning hech bo'lmaganda bittasi ∂D_1 ga tegishli bo'lishi uchun $z = 0$ yoki $\zeta = 0$ bo'lishi shart. Bundan esa $r = r_1$ bo'ladi, ya'ni $g(x, y, z; \xi, \eta, \zeta) = -1/r_1$. Bu funksiya D_1 sohada garmonikdir. Shunday qilib,

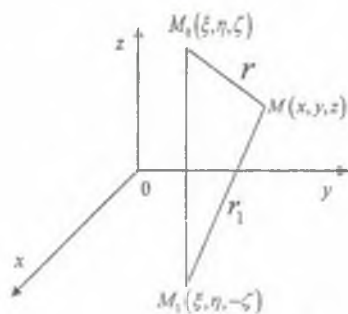
$$G(x, y; \xi, \eta) = \frac{1}{4\pi} \left(\frac{1}{r} + g(x, y, z; \xi, \eta, \zeta) \right) = \frac{1}{4\pi} \left(\frac{1}{r} - \frac{1}{r_1} \right)$$

funksiya D_1 sohada Laplas tenglamasining Grin funksiyasi bo'ladi.

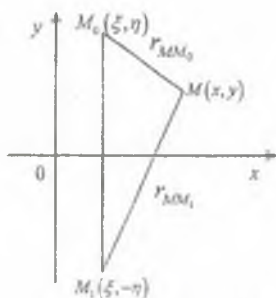
$\Delta u(x, y) = 0$ tenglamaning D_2 : $y > 0$ yarim tekislikdagi Grin funksiyasi quyidagi ko'rinishda bo'ladi:

$$G(x, y; \xi, \eta) = G(\xi, \eta; x, y) = \frac{1}{2\pi} \left[\ln \frac{1}{r_{MM_0}} - \ln \frac{1}{r_{MM_1}} \right], \quad (4.67)$$

bu yerda
$$\frac{r_{MM_0}}{r_{MM_1}} = \sqrt{(x - \xi)^2 + (y - \eta)^2} \quad (4.3\text{-chizma}).$$



4.2 - chizma



4.3 - chizma

Integral ifoda [11], [15]. Agar $u(x)$ funktsiya $C^2(D) \cap C^1(\bar{D})$ sinfga tegishli bo'lsa, u holda bu funksiyaning integral ifodasi

$$u(x) = \frac{1}{2\pi} \int_S \left[\frac{\partial u(\xi)}{\partial n} E(x, \xi) - u(\xi) \frac{\partial E(x, \xi)}{\partial n} \right] d_\xi S - \frac{1}{2\pi} \int_D E(x, \xi) \Delta u(\xi) d\xi =$$

$$= \begin{cases} \frac{1}{2\pi} \int_S \left[\frac{\partial u(\xi)}{\partial n} \ln \frac{1}{r} - u(\xi) \frac{\partial}{\partial n} \ln \frac{1}{r} \right] d_\xi S - \\ - \frac{1}{2\pi} \int_D \ln \frac{1}{r} \Delta u(\xi) d\xi, & n=2, \end{cases} \quad (4.68)$$

$$= \begin{cases} \frac{1}{(n-2)|S_1|} \int_S \left[\frac{1}{r^{n-2}} \frac{\partial u(\xi)}{\partial n} - u(\xi) \frac{\partial}{\partial n} \left(\frac{1}{r^{n-2}} \right) \right] d_\xi S - \\ - \int_D \frac{\Delta u(\xi)}{r^{n-2}} d\xi, & n>2 \end{cases} \quad (4.69)$$

ko'rinishda bo'ladi, bu yerda S_1 - birlik sfera.

Agar $u(x)$ funksiya D sohada garmonik funksiya bo'lsa, u holda (4.68) va (4.69) formulalarda $\Delta u = 0$ bo'lib, bu **garmonik funksiyaning integral ifodasi**

$$u(x) = \begin{cases} \frac{1}{2\pi} \int_S \left[\frac{\partial u(\xi)}{\partial n} \ln \frac{1}{r} - u(\xi) \frac{\partial}{\partial n} \ln \frac{1}{r} \right] d_\xi S, & n=2, \end{cases} \quad (4.70)$$

$$= \begin{cases} \frac{1}{(n-2)|S_1|} \int_S \left[\frac{1}{r^{n-2}} \frac{\partial u(\xi)}{\partial n} - u(\xi) \frac{\partial}{\partial n} \left(\frac{1}{r^{n-2}} \right) \right] d_\xi S, & n>2 \end{cases} \quad (4.71)$$

ko'rinishda bo'ladi.

Yuqoridagi integral ifodalarga o'xshash ushbu $\Delta u(x, y) = -f(x)$ Puasson tenglamasi uchun

$$u|_S = \varphi(x), \quad x \in S$$

Dirixle masalasining yechimi

$$u(x) = - \int_S \frac{\partial G}{\partial n_y} \cdot \varphi(y) dS_y + \int_D G(x, y) f(y) dy, \quad (4.72)$$

formula orqali topiladi, bu yerda $G(x, \xi)$ Grin funksiya bo'lib, (4.56) formuladan aniqlanadi.

Grin funksiyasi yordamida chegaraviy masalalarni yechish. Yarim tekislikda (4.2) Laplas tenglamasi uchun ($n=2$) Dirixle masalasini qaraymiz:

$$\Delta u \equiv \Delta u(x, y) = 0, \quad y > 0, \quad -\infty < x < +\infty, \quad (4.73)$$

$$u(x, 0) = f(x), \quad -\infty < x < +\infty. \quad (4.74)$$

(4.73), (4.74) masalaning yechimi quyidagi ko'rinishda

$$u(x, y) = \int_{-\infty}^{+\infty} f(\xi) \cdot \frac{\partial G(x, y, \xi, \eta)}{\partial \eta} \Big|_{\eta=0} d\xi \quad (4.75)$$

ifodalanadi, bu yerda $G(x, y, \xi, \eta)$ - funksiya (4.67) formula orqali aniqlanadi, ya'ni

$$G(x, y, \xi, \eta) = \frac{1}{2\pi} \left[\ln \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2}} - \ln \frac{1}{\sqrt{(x-\xi)^2 + (y+\eta)^2}} \right].$$

Bu funksiyaning η bo'yicha differensiallab, quyidagiga ega bo'lamiz:

$$\frac{\partial G}{\partial \eta} = -\frac{1}{2\pi} \cdot \frac{-(y-\eta)}{(x-\xi)^2 + (y-\eta)^2} + \frac{1}{2\pi} \cdot \frac{(y+\eta)}{(x-\xi)^2 + (y+\eta)^2}. \quad (4.76)$$

(4.76) ni (4.75) ga qo'yib, (4.73), (4.74) masalaning yechimini

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{+\infty} \frac{f(\xi)}{(x-\xi)^2 + y^2} d\xi \quad (4.77)$$

olamiz.

3-masala. Quyidagi

$$\Delta u(x, y) = 0, \quad y > 0, \quad -\infty < x < +\infty, \quad (4.73)$$

$$u(x, 0) = \frac{x}{x^2 + 1}, \quad -\infty < x < +\infty. \quad (4.78)$$

Dirixle masalasini yeching.

Yechish. (4.78) ni (4.77) ga qo'yib, quyidagini hisoblaymiz:

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{+\infty} \frac{\xi d\xi}{[(x-\xi)^2 + y^2](1+\xi^2)} = 2iy [\operatorname{res} f(i) + \operatorname{res} f(x+iy)], \quad (4.79)$$

bu yerda $f(z) = \xi / [(x-\xi)^2 + y^2](1+\xi^2)$.

Bundan, chegirmalar haqidagi teoremdan foydalanib, [9], [14]

$$\operatorname{res} f(i) = \frac{1}{2[(i-x)^2 + y^2]}, \quad \operatorname{res} f(x+iy) = \frac{x+iy}{2iy[1-(x+iy)^2]} \quad (4.80)$$

ni topamiz.

Buni (4.79) ga qo'yamiz:

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{+\infty} \frac{\xi d\xi}{[(x-\xi)^2 + y^2](1+\xi^2)} = \frac{iy}{(i-x)^2 + y^2} + \frac{x+iy}{1-(x+iy)^2} =$$

$$\begin{aligned}
&= \frac{iy}{[(i-x)+iy][(i-x)-iy]} + \frac{x+iy}{[(x+iy)-1][(x+iy)+1]} = \\
&= \frac{1}{2} \left[\frac{1}{i-x-iy} - \frac{1}{i-x+iy} \right] + \frac{1}{2} \left[\frac{1}{x+iy-1} - \frac{1}{1+x+iy} \right] = \\
&= \frac{1}{2} \left[-\frac{1}{i(y-1)+x} - \frac{1}{i(1+y)-x} + \frac{1}{x+i(y-1)} + \frac{1}{x+i(y+1)} \right] = \\
&= \frac{1}{2} \left[-\frac{1}{i(1+y)-x} + \frac{1}{x+i(y+1)} \right] = \frac{1}{2} \left[\frac{x-i(y+1)}{(1+y)^2+x^2} + \frac{x+i(y+1)}{x^2+(y+1)^2} \right] = \\
&= \frac{x}{x^2+(1+y)^2}.
\end{aligned}$$

Shunday qilib, (4.73), (4.78) masalaning yechimi

$$u(x, y) = \frac{x}{x^2 + (1+y)^2}$$

ko'rinishda bo'ladi.

Yarim fazoda (4.2) Laplas tenglamasi uchun ($n=3$) Dirixle masalasini qaraymiz:

$$\Delta u = \Delta u(x, y, z) = 0, \quad -\infty < x, y < +\infty, \quad z > 0,$$

$$u(x, y, 0) = f(x, y), \quad -\infty < x, y < +\infty.$$

Bu masalaning yechimi quyidagi ko'rinishda

$$u(x, y, z) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\xi, \eta) \frac{\partial G}{\partial \zeta} \Big|_{\zeta=0} d\xi d\eta$$

ifodalanadi, bu yerda $G(x, y, z; \xi, \eta, \zeta)$ — funksiya ushbu formula orqali aniqlanadi (4-§ dagi 2-masalaga qarang)

$$G(x, y, \zeta, \eta) = \frac{1}{4\pi} \left[\frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}} - \frac{1}{\sqrt{(x-\xi)^2 + (y+\eta)^2 + (z+\zeta)^2}} \right].$$

Bu funksiyaning ζ bo'yicha differensiallab, so'ng $\zeta=0$ qo'yib, quyidagiga ega bo'lamiz:

$$\frac{\partial G}{\partial \zeta} \Big|_{\zeta=0} = \frac{1}{4\pi} \cdot \frac{2z}{[(x-\xi)^2 + (y-\eta)^2 + z^2]^{3/2}}.$$

Demak, yarim fazoda ($z > 0$) (4.2) Laplas tenglamasi uchun Dirixle masalasining yechimi

$$u(x, y, z) = \frac{1}{4\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{2z}{[(x-\xi)^2 + (y-\eta)^2 + z^2]^{3/2}} f(\xi, \eta) d\xi d\eta \quad (4.80)$$

ko'rinishda bo'ladi.

4-masala. Quyidagi

$$\Delta u(x, y, z) = 0, \quad -\infty < x, y < +\infty, \quad z > 0,$$

$$u(x, y, 0) = \cos x \cdot \cos y, \quad -\infty < x, y < +\infty.$$

Dirixle masalasini yeching.

Yechish. Qo'yilgan Dirixle masalasining yechimi quyidagi

$$u(x, y, z) = \frac{z}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos \xi \cdot \cos \eta d\xi d\eta}{[(\xi-x)^2 + (\eta-y)^2 + z^2]^{3/2}}$$

formuladan topiladi. Bu integralni hisoblash uchun $\xi - x = w$, $\eta - y = v$ almashtirishlarni bajarib,

$$\begin{aligned} u(x, y, z) &= \frac{z}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos(x+w) \cdot \cos(y+v) dw dv}{[w^2 + v^2 + z^2]^{3/2}} = \\ &= \frac{z}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos w \cdot \cos x - \sin w \cdot \sin x}{[w^2 + v^2 + z^2]^{3/2}} [\cos v \cdot \cos y - \sin v \cdot \sin y] dw dv = \\ &= \frac{z}{2\pi} \cos x \cdot \cos y \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos w \cdot \cos v dw dv}{[w^2 + v^2 + z^2]^{3/2}}, \end{aligned} \quad (4.81)$$

ni hosil qilamiz, ifodadagi qolgan uchta integral, integral ostidagi funksiyalarning toqligiga ko'ra nolga teng bo'ladi.

Endi ushbu integralni hisoblaymiz:

$$\begin{aligned} J &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos w \cdot \cos v dw dv}{[w^2 + v^2 + z^2]^{3/2}} = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos(w+v) + \sin w \cdot \sin v dw dv}{[w^2 + v^2 + z^2]^{3/2}} = \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos(w+v) dw dv}{[w^2 + v^2 + z^2]^{3/2}}, \end{aligned}$$

chunki

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\sin w \cdot \sin v dw dv}{[w^2 + v^2 + z^2]^{3/2}} = 0$$

Oxirgi integralda $\mu = \frac{1}{\sqrt{2}}(w+v)$, $\rho = \frac{1}{\sqrt{2}}(w-v)$ almashtirishlarni bajarib,

$$\begin{aligned}
 J &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\cos(\sqrt{2}\mu) d\mu d\rho}{[\mu^2 + \rho^2 + z^2]^{3/2}} = \int_{-\infty}^{+\infty} \cos(\sqrt{2}\mu) d\mu \int_{-\infty}^{+\infty} \frac{d\rho}{[\mu^2 + \rho^2 + z^2]^{3/2}} = \\
 &= \left\{ \rho = \sqrt{\mu^2 + z^2} \operatorname{tg} t \right\} = \int_{-\infty}^{+\infty} \cos(\sqrt{2}\mu) d\mu \int_{-\pi/2}^{\pi/2} \frac{|\cos t| dt}{\mu^2 + z^2} = \\
 &= 2 \int_{-\infty}^{+\infty} \frac{\cos(\sqrt{2}\mu)}{\mu^2 + z^2} d\mu \quad (4.82)
 \end{aligned}$$

ga ega bo'lamiz.

(4.82) formulaga chegirmalar haqidagi Koshi [14] teoremasini qo'llaymiz:

$$\begin{aligned}
 J &= \operatorname{Re} \int_{-\infty}^{+\infty} \frac{(e^{i\sqrt{2}\mu} + e^{-i\sqrt{2}\mu})}{\mu^2 + z^2} d\mu = 2 \operatorname{Re} \int_0^{+\infty} \frac{e^{i\sqrt{2}\mu}}{\mu^2 + z^2} d\mu = 4\pi i \operatorname{res} \frac{e^{i\sqrt{2}\mu}}{\mu^2 + z^2} \Big|_{\mu=zi} = \\
 &= 4\pi i \cdot \frac{e^{-z\sqrt{2}}}{2zi} = \frac{2\pi}{z} \cdot e^{-z\sqrt{2}}.
 \end{aligned}$$

Bundan va (4.81)dan qo'yilgan Dirixle masalasining yechimini

$$u(x, y, z) = e^{-z\sqrt{2}} \cos x \cdot \cos y.$$

ko'rinishda topamiz.

Boshqa usullar yordamida chegaraviy masalalarni yechish

1- izoh. (4.73), (4.78) masalaning yechimini Grin funksiyasidan foydalanmay ham topish mumkin. Haqiqatan,

$$u(x, y) = \frac{1}{2\pi} \ln \frac{1}{\sqrt{x^2 + (y + \alpha)^2}} \quad (\alpha > 0)$$

funksiya yuqori yarim tekislikda ($y > 0$) (4.73) Laplas tenglamasini qanoatlantiradi, ya'ni

$$\Delta u(x, y) = \Delta \ln \frac{1}{\sqrt{x^2 + (y + \alpha)^2}} = 0.$$

Bundan tashqari, quyidagi tengliklar o'rinni:

$$\frac{\partial}{\partial x} \Delta \ln \frac{1}{\sqrt{x^2 + (y + \alpha)^2}} = 0 \quad \text{yoki} \quad \Delta \frac{\partial}{\partial x} \ln \frac{1}{\sqrt{x^2 + (y + \alpha)^2}} = 0,$$

ya'ni

$$\Delta\left(\frac{x}{r^2}\right)=0, \quad r^2=x^2+(y+\alpha)^2.$$

Shunday qilib, $u(x,y)=\frac{x}{x^2+(y+\alpha)^2}$ funksiya yuqori yarim tekislikda garmonikdir. Bundan, (4.78) chegaraviy shartga ko'ra $\alpha=1$ ga teng bo'lib, $u(x,y)=\frac{x}{x^2+(1+y)^2}$ funksiya (4.73), (4.78) masalaning yechimi bo'ladi.

2- izoh. Agar $z=x+iy$, $\frac{y}{(x-\xi)^2+y^2}=\operatorname{Re}\frac{1}{i(\xi-z)}$ bo'lsa, u holda, (4.77) formulani quyidagi ko'rinishda

$$u(z)=\operatorname{Re}\frac{1}{\pi i}\int_{-\infty}^{+\infty}\frac{u(\xi)}{\xi-z}d\xi \quad (4.83)$$

yozib olamiz.

Endi yarim tekislikda ($\operatorname{Im} z>0$) (ya'ni $y>0$ da) Laplas tenglamasi uchun Dirixle masalasini ko'raylik:

$$\Delta u(x,y)=0, \quad -\infty<x<+\infty, \quad y>0,$$

$$u(x,y)\Big|_{y=0}=R(x), \quad -\infty<x<+\infty,$$

bu yerda $R(z)$ haqiqiy o'qda qutb maxsus nuqtaga ega bo'lmagan haqiqiy ratsional funksiya bo'lib, $z\rightarrow\infty$ da $R(z)\rightarrow 0$.

Berilgan Dirixle masalasining yechimi

$$u(z)=\operatorname{Re}\frac{1}{\pi i}\int_{-\infty}^{+\infty}\frac{R(\xi)}{\xi-z}d\xi \quad (4.84)$$

formula orqali topiladi. (4.84) formuladagi integral chegirmalar haqidagi Koshi [9], [14] teoremasi yordamida hisoblanadi:

$$u(z)=-2\operatorname{Re}\sum_{\operatorname{Im}\zeta_0<0}\operatorname{res}_{\zeta=\zeta_0}\frac{R(\zeta)}{\zeta-z}. \quad (4.85)$$

5-masala. Quyidagi $\Delta u(x,y)=0$, $-\infty<x<+\infty$, $y>0$,

$$u(x,0)=\frac{k}{x^2+1}, \quad -\infty<x<+\infty, \quad k=\operatorname{const}$$

Dirixle masalasini yeching.

Yechish. (4.84) va (4.85) formulalardan foydalanib, Dirixle masalasining yechimini quyidagi ko'rinishda topamiz:

$$u(z) = -2 \operatorname{Re}_{\zeta=-i} \frac{k}{(\zeta-z)(1+\zeta^2)} = -2 \operatorname{Re} \frac{k}{2i(z+i)} = \frac{k(y+1)}{x^2+(y+1)^2}.$$

6-masala. Quyidagi

$$\Delta u(x, y, z) = z e^{-z} \sin x \cdot \sin y, \quad -\infty < x, y < +\infty, \quad z > 0,$$

$$u(x, y, 0) = 0, \quad -\infty < x, y < +\infty$$

masalaning $z \rightarrow +\infty$ da chegaralangan $u(x, y, z)$ yechimini toping.

Yechish. Berilgan masala yechimini

$$u(x, y, z) = \mathcal{G}(z) \cdot \sin x \cdot \sin y$$

ko'rinishda izlaymiz. U holda

$$\begin{aligned} \Delta u \equiv \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} &= -\mathcal{G}(z) \cdot \sin x \cdot \sin y - \mathcal{G}(z) \cdot \sin x \cdot \sin y + \\ &+ \mathcal{G}''(z) \cdot \sin x \cdot \sin y \end{aligned}$$

ni hisobga olib, quyidagi tenglikka

$$-2\mathcal{G}(z) \cdot \sin x \cdot \sin y + \mathcal{G}''(z) \cdot \sin x \cdot \sin y = z e^{-z} \sin x \cdot \sin y$$

ega bo'lamiz. Bundan quyidagi

$$\mathcal{G}''(z) - 2\mathcal{G}(z) = z e^{-z}$$

oddiy differensial tenglamani hosil qilamiz. Uning umumiy yechimi

$$\mathcal{G}(z) = c_1 e^{\sqrt{2}z} + c_2 e^{-\sqrt{2}z} + e^{-z} (2-z) \quad (4.86)$$

ko'rinishda bo'ladi.

Endi c_1 va c_2 o'zgarmaslarni chegaraviy shartlar yordamida topamiz. (4.86) yechimda $c_1 = 0$, chunki agar $c_1 \neq 0$ bo'lsa, u holda $z \rightarrow +\infty$ da $\mathcal{G}(z) \rightarrow \infty$ bo'ladi. Shuning uchun (4.86) yechimda $z = 0$ deb, $0 = \mathcal{G}(0) = c_2 + 2 \Rightarrow c_2 = -2$ aniqlaymiz. Demak, (4.86) ga ko'ra qo'yilgan masalaning yechimi

$$u(x, y, z) = \left[e^{-z} (2-z) - 2e^{-\sqrt{2}z} \right] \sin x \cdot \sin y.$$

ko'rinishda topamiz.

7-masala. Quyidagi $\Delta u(x, y, z) = 0, \quad -\infty < x, y < +\infty, \quad z > 0,$

$$u(x, y, 0) = \frac{x^2 + y^2 - 2}{(x^2 + y^2 + 1)^{5/2}}, \quad -\infty < x, y < +\infty$$

masalaning $u(x, y, z)$ yechimini toping.

Yechish. Shuni aytish mumkinki, $z > 0$ yarim fazoning hamma joyida $u(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + (1+z)^2}}$ funksiya $u_{xx} + u_{yy} + u_{zz} = 0$ Laplas tenglamasini qanoatlantiradi, ya'ni

$$\Delta \left[\frac{1}{\sqrt{x^2 + y^2 + (1+z)^2}} \right] = 0.$$

Bu tenglikni z bo'yicha ikki marta differensiallab, quyidagiga

$$\Delta \left[\frac{x^2 + y^2 - 2(1+z)^2}{(x^2 + y^2 + (1+z)^2)^{5/2}} \right] = 0 \quad (4.87)$$

ega bo'lamiz. Bundan xulosa qilib, shuni aytish lozimki

$$u(x, y, z) = \frac{x^2 + y^2 - 2(1+z)^2}{(x^2 + y^2 + (1+z)^2)^{5/2}} \quad (4.88)$$

funksiya $z > 0$ yarim fazoning hamma joyida garmonik bo'lib ((4.87) ga qarang), $z = 0$ da quyidagi

$$u|_{z=0} = \frac{x^2 + y^2 - 2}{(x^2 + y^2 + 1)^{5/2}}$$

chegaraviy shartni qanoatlantiradi.

Shunday qilib, quyilgan masalaning yechimi (4.88) formula orqali topiladi.

Bigarmonik tenglama. Ushbu $\Delta^2 u = 0$ bigarmonik va $\Delta^2 u = f$ tenglamalar uchun Grin funksiyasidan foydalanmay, chegaraviy masalalarni o'rganish mumkin.

8-masala. Quyidagi $K = \{(r, \varphi) : 0 \leq r \leq a, 0 \leq \varphi < 2\pi\}$ sohada

$$\begin{aligned} \Delta^2 u &= 0 \quad K \text{ da}, \\ u|_{r=a} &= 0, \quad \left. \frac{\partial u}{\partial n} \right|_{r=a} = A \cos \varphi \quad \partial K \text{ da} \end{aligned}$$

masalani yeching, n - doiraning chegarasida o'tkazilgan tashqi birlik normal.

Yechish. Bizga ma'lumki, bigarmonik tenglama uchun Dirixle masalasi:

$$u|_{r=a} = \psi, \quad \frac{\partial u}{\partial n} \bigg|_{r=a} = \phi$$

ushbu ko'rinishdagi yechimga ega

$$u(r, \varphi) = \frac{1}{2a\pi} (r^2 - a^2)^2 \left[-\frac{1}{2} \int_0^{2\pi} \frac{\phi(t) dt}{r^2 + a^2 - 2ar \cos(\varphi - t)} + \int_0^{2\pi} \frac{\psi(t) [a - r \cos(\varphi - t)] dt}{[r^2 + a^2 - 2ar \cos(\varphi - t)]^{\frac{3}{2}}} \right], \quad (4.89)$$

(4.89) formulaga $\psi = 0$, $\phi = A \cos \varphi$ qo'yib,

$$u(r, \varphi) = -\frac{A}{2a} (r^2 - a^2)^2 \cdot \int_0^{2\pi} \frac{1}{2\pi} \cdot \frac{\cos t dt}{r^2 + a^2 - 2ar \cos(\varphi - t)}$$

ni hosil qilamiz. IV bobning 3-§iga ko'ra oxirgi integral Laplas tenglamasi uchun qo'yilgan

$$\Delta v = 0 \quad K \partial a,$$

$$v|_{r=a} = \cos \varphi \quad \partial K \partial a$$

Dirixle masalasining yechimidir. Bu masalaning yechimi $v = r/a \cos \varphi$ ko'rinishda bo'ladi. Demak, qo'yilgan masala yechimining yagonaligiga ko'ra qo'yidagi tenglikka

$$\frac{1}{2\pi} \cdot \int_0^{2\pi} \frac{\cos t dt}{r^2 + a^2 - 2ar \cos(\varphi - t)} = \frac{r}{a} \cos \varphi$$

ega bo'lamiz.

Shunday qilib, bigarmonik tenglama uchun qo'yilgan Dirixle masalasining yechimi

$$u(r, \varphi) = -\frac{Ar(r^2 - a^2)^2}{2a^2} \cos \varphi$$

ko'rinishda bo'ladi.

9-masala. Quyidagi $K = \{(r, \varphi): 0 \leq r \leq a, 0 \leq \varphi < 2\pi\}$ sohada

$$\Delta^2 u = 1 \quad K \partial a,$$

$$u|_{r=a} = 0, \quad \frac{\partial u}{\partial n} \bigg|_{r=a} = 0, \quad \partial K \partial a$$

masalani yeching, n - doiraning chegarasida o'tkazilgan tashqi birlik

normal.

Yechish. Faraz qilaylik, izlanayotgan yechim faqat r ga bog'liq bo'lsin, ya'ni $u = u(r)$. U holda

$$\Delta^2 u = \left(\frac{\partial^2}{\partial r^2} + \frac{\partial}{r \partial r} \right)^2 u = \left(\frac{\partial^4}{\partial r^4} + \frac{2\partial^3}{r \partial r^3} + \frac{\partial^2}{r^2 \partial r^2} \right) u$$

o'rinlidir.

Shunday qilib, oddiy differensial tenglama uchun quyidagi masalaga kelamiz:

$$\frac{d^4 u}{dr^4} + \frac{2d^3 u}{r dr^3} + \frac{d^2 u}{r^2 dr^2} = 1, \quad 0 < r < a, \quad u|_{r=a} = 0, \quad \frac{\partial u}{\partial r}|_{r=a} = 0.$$

Bu masalani yechib, qo'yilgan masalaning yechimini

$$u(r) = \frac{a^4}{7} \left[\frac{1}{12} \left(\frac{r}{a} \right)^4 - \frac{1}{3} \left(\frac{r}{a} \right) + \frac{1}{4} \right]$$

ko'rinishda topamiz.

10-masala. Quyidagi $E = \{ (x, y) : -\infty < x < +\infty, y > 0 \}$ yarim

tekislikda $\Delta^2 u = e^{-2y} \sin x$ E da $u|_{y=0} = 0, \frac{\partial u}{\partial y}|_{y=0} = 0, -\infty < x < +\infty$

masalani yeching.

Yechish. Berilgan tenglamani quyidagi ko'rinishda yozib olamiz:

$$\frac{\partial^4 u}{\partial x^4} + 2 \frac{\partial^2 u}{\partial x^2 \partial y^2} + \frac{\partial^4 u}{\partial y^4} = e^{-2y} \sin x.$$

Bu tenglamaning yechimini $u(x, y) = \omega(y) \sin x$ ko'rinishda izlaymiz, $\omega(y)$ — noma'lum funksiya. $u(x, y) = \omega(y) \sin x$ funksiyaning berilgan tenglamaga qo'yib,

$$\omega^{(IV)} - 2\omega'' + \omega = e^{-2y}$$

oddiy differensial tenglamani olamiz. Uning umumiy yechimi

$$\omega(y) = C_1 e^y + C_2 y e^y + C_3 e^{-y} + C_4 y e^{-y} + \frac{1}{9} e^{-2y}$$

ko'rinishda bo'ladi. Masala shartiga ko'ra $C_1 = 0, C_2 = 0$ teng, aksincha, $y \rightarrow +\infty$ da $\omega(y) \rightarrow \infty$ bo'ladi. C_3, C_4 lar $\omega(0) = \omega'(0) = 0$ shartlardan topiladi. Demak, oddiy differensial

tenglamaning yechimi

$$\omega(y) = \frac{1}{9} \left(-e^{-y} + y e^{-y} + e^{-2y} \right)$$

ko'rinishda bo'lib, qo'yilgan masalasining yechimi esa

$$u(x, y) = \frac{1}{9} \left(-e^{-y} + y e^{-y} + e^{-2y} \right) \sin x$$

formuladan topiladi.

Mustaqil yechish uchun masalalar

593. $\Delta u(x, y, z) = 0$ tenglamaning $y > 0, z > 0$ ikki yoqli

burchakdagi Grin funksiyasi $G(x, y, z; \xi, \eta, \zeta) = \frac{1}{4\pi} \left(\frac{1}{r} - \frac{1}{r_1} - \frac{1}{r_2} + \frac{1}{r_3} \right)$

ko'rinishga ega bo'lishini ko'rsating. Bu yerda

$$r = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}, \quad r_1 = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z+\zeta)^2},$$

$$r_2 = \sqrt{(x-\xi)^2 + (y+\eta)^2 + (z-\zeta)^2}, \quad r_3 = \sqrt{(x-\xi)^2 + (y+\eta)^2 + (z+\zeta)^2}.$$

594. $\Delta u(x, y, z) = 0$ tenglamaning $x > 0, y > 0, z > 0$ oktantadagi

$$\text{Grin funksiyasi } G(x, y, z; \xi, \eta, \zeta) = \frac{1}{4\pi} \sum_{k=0}^7 (-1)^k \frac{1}{r_k}$$

ko'rinishga ega bo'lishini ko'rsating. Bu yerda

$$r_0 = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}, \quad r_1 = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z+\zeta)^2},$$

$$r_2 = \sqrt{(x-\xi)^2 + (y+\eta)^2 + (z+\zeta)^2}, \quad r_3 = \sqrt{(x-\xi)^2 + (y+\eta)^2 + (z-\zeta)^2},$$

$$r_4 = \sqrt{(x+\xi)^2 + (y+\eta)^2 + (z-\zeta)^2}, \quad r_5 = \sqrt{(x+\xi)^2 + (y-\eta)^2 + (z-\zeta)^2},$$

$$r_6 = \sqrt{(x+\xi)^2 + (y-\eta)^2 + (z+\zeta)^2}, \quad r_7 = \sqrt{(x+\xi)^2 + (y+\eta)^2 + (z+\zeta)^2}.$$

595. $\Delta u = 0$ tenglamaning radiusi $R = 4$ ga teng bo'lgan shardagi

$$\text{Grin funksiyasi } G(x, y) = \frac{1}{4\pi} \left(\frac{1}{|x-y|} - \frac{4}{|y||x-y^*|} \right)$$

ko'rinishga ega bo'lishini ko'rsating.

596. $\Delta u(x, y, z) = 0$ tenglamaning ikkita $z = 0$ va $z = l$ tekisliklar bilan chegaralangan sohadagi Grin funksiyasi

$$G(x, y, z; \xi, \eta, \zeta) = \frac{1}{4\pi} \sum_{n=-\infty}^{\infty} \left(\frac{1}{r_n} - \frac{1}{r'_n} \right)$$

ko'rinishga ega bo'lishini ko'rsating, bu yerda

$$r_n = \sqrt{(x-\xi)^2 + (y-\eta)^2 + [z - (2nl + \zeta)]^2},$$

$$r'_n = \sqrt{(x-\xi)^2 + (y-\eta)^2 + [z - (2nl - \zeta)]^2}.$$

597. $\Delta u(x, y, z) = 0$ tenglamaning $z > 0$ yarim fazoda $z = 0$ da quyidagi $\frac{\partial u}{\partial z} + hu = 0$ chegaraviy shartni qanoatlantiradigan Grin funksiyasini toping.

598. $\Delta u(x, y, z) = 0$ tenglamaning quyidagi sohadagi Grin funksiyasini toping:

- 1) $x^2 + y^2 + z^2 < R^2$, $z > 0$ yarim sharda;
- 2) $x^2 + y^2 + z^2 < R^2$, $y > 0$ $z > 0$ sharning to'rtidan bir qismida;
- 3) $x^2 + y^2 + z^2 < R^2$, $x > 0$ $y > 0$ $z > 0$ sharning sakkizdan bir qismida.

599. $\Delta u(z) = 0$, $z = x + iy$ tenglamaning quyidagi sohadagi Grin funksiyasini toping:

- 1) $\text{Im } z > 0$ yarim tekislikda; 2) $0 < \arg z < 0,5\pi$ tekislikning to'rtidan birida;
- 3) $|z| < R$ doirada; 4) $|z| < R$ $\text{Im } z > 0$ yarim doirada;
- 5) $|z| < 1$, $0 < \arg z < 0,5\pi$ doiraning to'rtidan birida;
- 6) $0 < \text{Im } z < \pi$ "polasa"da;
- 7) $0 < \text{Im } z < \pi$, $\text{Re } z > 0$ yarim polasada;

600. $D = \{(x, y): -1 < x < 1, 0 < y < \sqrt{1-x^2}\}$ sohada

$$\begin{cases} \Delta u(x, y) = 0, & -1 < x < 1, & 0 < y < \sqrt{1-x^2}, \\ u|_{x^2+y^2=r^2} = \varphi(s), & 0 \leq s \leq l, & \frac{\partial u}{\partial y}|_{y=0} = v(x) & -1 < x < 1 \end{cases}$$

masalaning Grin funksiyasini tuzing va yechimni toping.

601. $D = \{(x, y): -1 < x < 1, 0 < y < \sqrt{1-x^2}\}$ sohada

$$\begin{cases} \Delta u(x, y) = 0, & -1 < x < 1, & 0 < y < \sqrt{1-x^2}, \\ u|_{x^2+y^2=r^2} = \varphi(s), & 0 \leq s \leq l, & u|_{y=0} = \tau(x), & -1 \leq x \leq 1 \end{cases}$$

masalaning Grin funksiyasini tuzing va yechimni toping.

$$602. D = \left\{ (x, y): -1 < x < 1, 0 < y < \left[\left(\frac{m+2}{2} \right)^2 (1-x^2) \right]^{1/(m+2)} \right\} \text{ sohada}$$

$$\begin{cases} y^m u_{xx} + u_{yy} = 0, & u|_{\Gamma} = \varphi(s), \quad 0 \leq s \leq l, \\ u|_{y=0} = \tau(x), \quad -1 \leq x \leq 1; \quad \Gamma: x^2 + \frac{4}{(m+2)^2} y^{m+2} = 1 \end{cases}$$

masalaning Grin funksiyasini tuzing va yechimni toping.

$$603. D = \left\{ (x, y): -1 < x < 1, 0 < y < \left[\left(\frac{m+2}{2} \right)^2 (1-x^2) \right]^{1/(m+2)} \right\} \text{ sohada}$$

$$\begin{cases} y^m u_{xx} + u_{yy} = 0, & u|_{\Gamma} = \varphi(s), \quad 0 \leq s \leq l, \\ \frac{\partial u}{\partial y} \Big|_{y=0} = \nu(x), \quad -1 < x < 1; \quad \Gamma: x^2 + \frac{4}{(m+2)^2} y^{m+2} = 1 \end{cases}$$

masalaning Grin funksiyasini tuzing va yechimini toping.

$$604. \text{ Ushbu } \Delta u = f(x, y, z), \quad -\infty < x, y < +\infty, \quad z > 0, \quad u|_{z=0} = u_0(x, y),$$

$-\infty < x, y < +\infty$ Dirixle masalasini f va u_0 quyidagi

qiymatlarida yeching. (f va u_0 - bo'lakli-uzluksiz va chegaralangan funksiyalar)

$$1) f = e^{-z} \sin x \cdot \cos y, \quad u_0 = 0; \quad 2) f = 0, \quad u_0 = \theta(y-x) = \begin{cases} 1, & y-x \geq 0, \\ 0, & y-x < 0; \end{cases}$$

$$3) f = 0, \quad u_0 = (1+x^2+y^2)^{-1/2}; \quad 4) f = 2((1+z)^2 + x^2 + y^2)^{-2}, \quad u_0 = (1+x^2+y^2)^{-1};$$

$$605. \text{ Ushbu } \Delta u(x) = 0, \quad x_2 > 0, \quad x_3 > 0, \quad u|_S = u_0(x), \quad x = (x_1, x_2, x_3)$$

Dirixle masalasini u_0 ning quyidagi qiymatlarida yeching:

1) u_0 - bo'lakli-uzluksiz va chegaralangan funksiya;

$$2) u_0|_{x_2=0} = 0, \quad u_0|_{x_3=0} = \exp\{-4x_1\} \cdot \sin 5x_2;$$

$$3) u_0|_{x_2=0} = 0, \quad u_0|_{x_3=0} = x_2(1+x_1^2+x_2^2)^{-3/2}.$$

606. Dirixle masalasini $|x| < R$ sharda yeching:

$$\Delta u(x) = -f(x), \quad |x| < R, \quad u|_{|x|=R} = \varphi(x), \quad |x| = \sqrt{x_1^2 + x_2^2 + x_3^2}$$